Sniper or Slump: Predicting NBA 3-Point Prop Bets

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Recording:
https://youtu.be/896XbHvkxMA

Problem Statement & Motivation

The Context: The "3-Point Revolution" in the NBA (2015–Present).

The Problem: "Prop Betting" (Proposition Bets) is the fastest-growing segment of sports gambling.

The Target: Can we predict if a player will make **3 or more** 3-pointers in a game?

Binary Classification: 1 (Hit Over) vs. 0 (Hit Under).

Data & Methodology

Data Source: NBA Game Data (2015 – 2022).

Training Set: Seasons 2015–2021 (~180,000 player-games).

Testing Set: Season 2022 (~28,000 player-games).

Tools Used: R (Tidyverse, Zoo, Ranger).

Feature Engineering

Recent Form: Rolling Average of 3s Made (Last 5 Games).

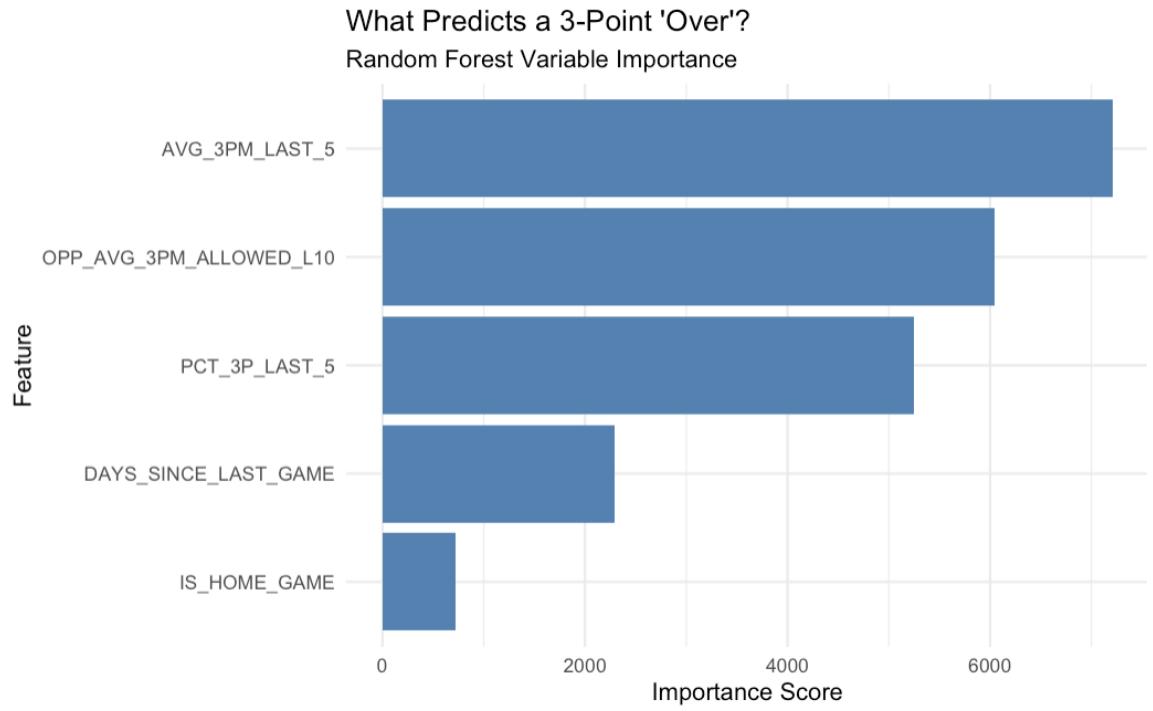
Shooting Efficiency: Rolling 3P% (Last 5 Games).

Fatigue: Days Since Last Game.

Opponent Defense: OPP_AVG_3PM_ALLOWED_L10 (How many 3s the opponent allows on average).

Context: Home vs. Away.

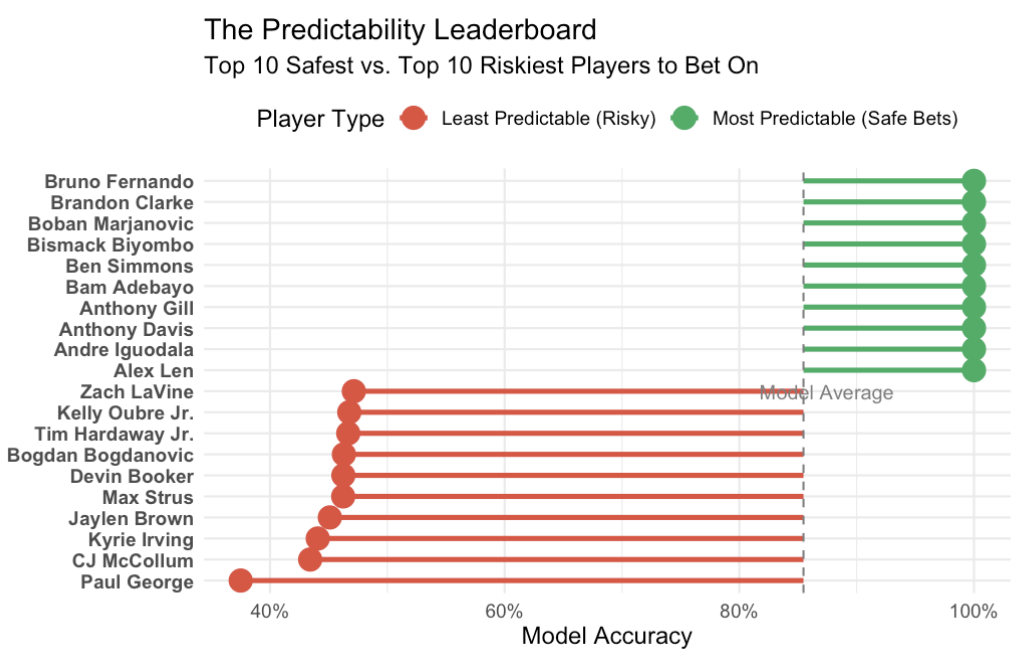
Exploratory Analysis - Variable Importance



Model Performance (Results)

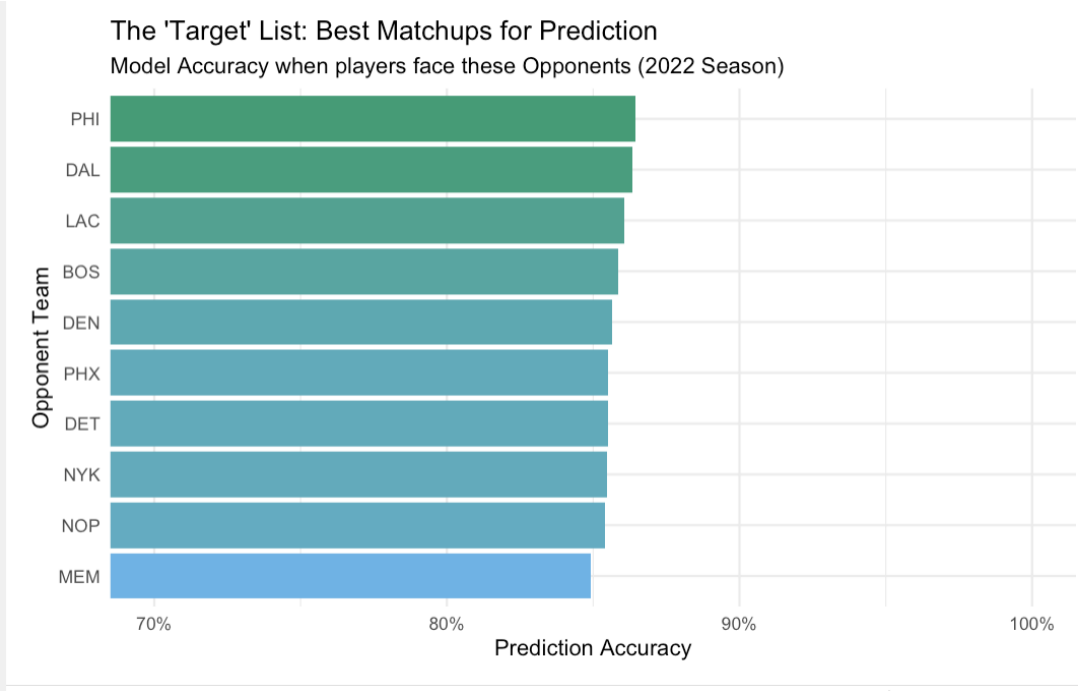
- **Model 1:** Logistic Regression
  - *Accuracy:* **84.23%**
- **Model 2:** Random Forest
  - *Accuracy:* **83.99%**
- **Key Insight:** Simpler models performed slightly better.
- **Confusion Matrix:**
  - Correctly Predicted "Under": ~22,800
  - Correctly Predicted "Over": ~929

Player Predictability Leaderboard: Risk vs. Reward



Strategic Insights - Target Matchups

- The "Target List": Best Matchups for Prediction
- **Bullet Points:**
  - **Top Targets:** PHI (76ers) & DAL (Mavericks).
  - **Accuracy:** >86.5% against these teams.
  - **Why?:** High defensive consistency = Predictable outcomes.



Case Study - The Superstars

- **Table/List of Star Player Accuracy:**
  - Giannis Antetokounmpo: **96%** (Easy - always Under)
  - Stephen Curry: **73.8%** (Impressive for a high-volume shooter)
  - LeBron James: **56.8%** (Difficult to predict)
  - Trae Young: **60.7%**

Conclusion & Recommendations

- **Summary:**
  - Recent form is the best predictor of future performance.
  - Overall Accuracy ~84%.
  - Star players vary wildly in predictability.
- **Actionable Insight:** Avoid betting on "Streaky" shooters (Paul George). Target consistent high-volume shooters (Curry) or non-shooters (Centers).

Future Work

- **Limitations:**
  - Did not account for injuries.
  - Did not use "Player Tracking" data (shot difficulty, defender distance).
- **Next Steps:**
  - Integrate defensive match-up data (Who is guarding the player?).
  - Create a "Risk Score" for every bet.